



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2014

Year 11 Yearly

Mathematics Continuers

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may NOT be awarded for messy or badly arranged work.
- **EACH SECTION IS TO BE RETURNED IN A SEPARATE BUNDLE**

Section A: Q1 and Q2.

Section B: Q3 and Q4.

- All necessary working should be shown in every question if full marks are to be awarded.
- Leave all answers in simplified exact form unless indicated otherwise.

Total Marks – 90

- Attempt all questions.
- All questions are not of equal value.

Examiner: *A Ward*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

SECTION A

Start Question 1 in a new answer booklet.

Question 1 (25 Marks)	Marks
a) What is the range of $y = \sqrt{4 - x^2}$?	1
b) Graph the solution of $ 2x + 3 \geq 2$ on a number line.	2
c) If $f(x) = 2ax + b$, find the values of a and b given that $f(-1) = 5$ and $f(2) = -1$	2
d) Differentiate with respect to x :	10
(i) $6x^5 - 3x^3 - 7x + 1$	
(ii) $\frac{7}{x}$	
(iii) $4(1 - x^3)^5$	
(iv) $\ln(6x^2 - 3)$	
(v) $\frac{x-1}{3x-4}$	
(vi) xe^x	
e) Find a primitive of:	6
(i) $(x^2 - 2)^2$	
(ii) $\frac{x^2 - 2}{x}$	
(iii) $\frac{x}{x^2 - 2}$	
f) Sketch the curves in separate diagrams, showing intercepts and asymptotes:	4
(i) $y = \log(x + 2)$	
(ii) $y = \frac{1}{x + 2}$	

End of Question 1.

Start Question 2 in a new answer booklet.

Question 2 (20 Marks)

Marks

a) Find the value(s) of x for which $\ln x = 2 \ln x$

1

b) If $f(x) = e^{x-2}$ find $f(-1)$ to 4 decimal places.

1

c) Find the value of the following:

6

(i) $\int_0^3 \sqrt{x} \, dx$

(ii) $\int_0^3 \frac{dx}{(2x-3)^2}$

(iii) $\int_1^{e-1} \frac{1}{x+1} \, dx$

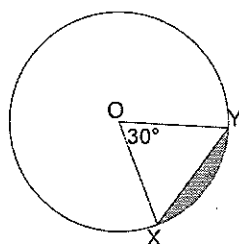
d) (i) Show that $g(x) = \frac{2}{x^2-1}$ is an even function.

3

(ii) State the domain of $y = g(x)$.

e) Calculate the exact shaded area where arc XY , of length 10π cm subtends an angle of 30° at, O the centre of the circle.

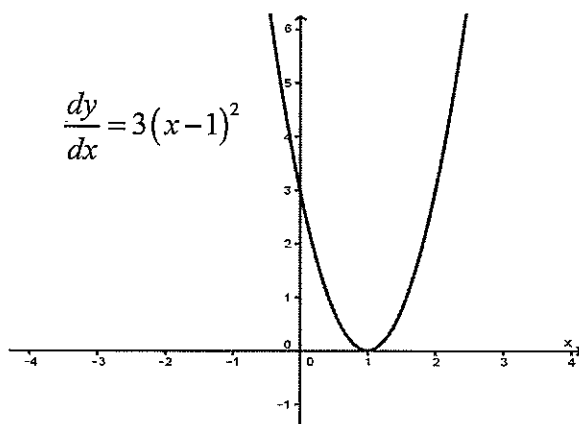
2



Not to scale

f) The gradient function of a curve is illustrated by the graph below.

7



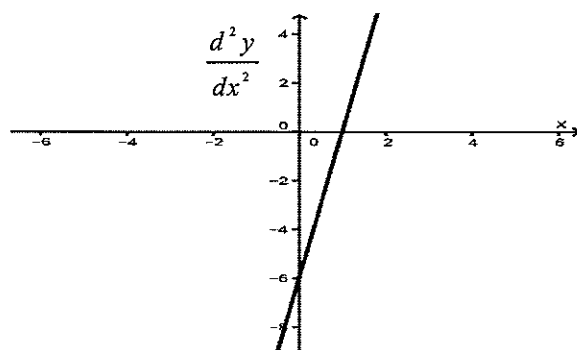
(i) A stationary point is located at $x = 1$.

A) Justify this statement with reference to the graph.

B) Comment on the sign of dy/dx for all x , $x \neq 1$.

C) What does this imply about the curve $y = f(x)$?

(ii) The graph of $\frac{d^2y}{dx^2}$, related to the above, is given below:



Copy and complete the table below:

x	0	1	2
Sign of $\frac{d^2y}{dx^2}$			

(iii) What is the nature of the stationary point at $x = 1$?

(iv) If the curve $y = f(x)$ passes through $(0, 0)$, find the equation of the curve.

End of Question 2

End of Section A

SECTION B

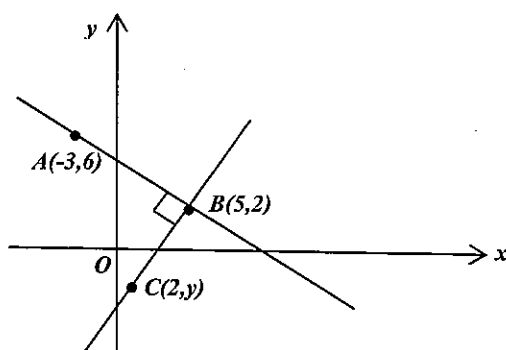
Start Question 3 in a new answer booklet

Question 3 (25 Marks)

Marks

- a) The diagram shows the origin O and the points $A(-3,6)$, $B(5,2)$ and $C(2,y)$.
The lines AB and BC are perpendicular.

11



NOT TO SCALE

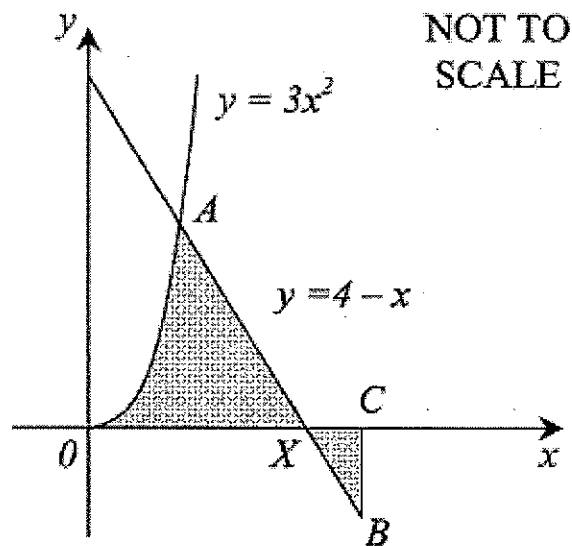
- (i) Copy this diagram into your writing booklet.
 - (ii) Show that A and B lie on the line $x + 2y = 9$.
 - (iii) Find the exact length AB .
 - (iv) Find the perpendicular distance from O to AB exactly.
 - (v) Find the area of triangle OAB .
 - (vi) Show that C has coordinates $(2, -4)$.
 - (vii) Does the line AC pass through the origin? Explain.
 - (viii) The point D is not shown on the diagram. The point D lies on the x -axis and $ABCD$ is a rectangle. Find the coordinates of D .
 - (ix) On your diagram, shade the region satisfying the inequality $x + 2y \geq 9$.
- b) Find the equation of the normal to the curve $y = x \log_e x$ at the point (e, e) . Give your answer in general form.
- c) For what value of x is the tangent to the curve $y = e^{3x}$ parallel to the line $y = 6x$?

3

3

- d) The shaded region $OABC$ is bounded by the lines $x = 0$, $x = 5$, the curve $y = 3x^2$, the line $y = 4 - x$ and the x -axis, as in the diagram below.

5



- (i) Find the co-ordinates of X and A .
- (ii) Find the area of the shaded region $OABC$.

- e) Differentiate $y = (\ln x)^2$ and hence evaluate $\int_1^2 \frac{\ln x}{x} dx$.

3

End of Question 3

Start Question 4 in a new answer booklet.

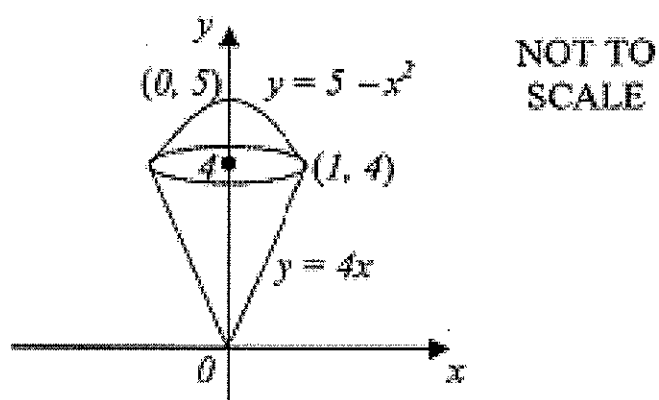
Question 4 (20 Marks)

Marks

- a) Given $\frac{d}{dx}(b^x) = b^x \log_e b$, evaluate $\int_0^\pi \pi^x dx$ correct to one decimal place. 2
- b) To calculate the area of the region bounded by the curve $y = x^2 - 3x$ and the x -axis between the ordinates of $x = 0$ and $x = 6$, a student used $y = \int_0^6 (x^2 - 3x) dx$. 3
- a) Explain why the student's method of calculating the area is incorrect.
- b) Find the area of the required region.
- c) Consider $f(x) = \frac{\log_e \sqrt{x}}{x}$. 5
- What is the domain of $f(x)$?
- (i) Find $f'(x)$ and hence determine all the stationary points.
- (ii) Sketch the curve of $y = f(x)$ clearly showing all its essential features.
- d) Differentiate $y = \log_2 x$. 2
- e) Determine the equation of the locus of the points $P(x, y)$ equidistant from the y -axis and the point $(1, 0)$. 3

- f) The diagram shows a cone and a paraboloid. It represents an ice-cream cone which is completely full of ice-cream and which has an additional scoop of ice-cream on top.

5



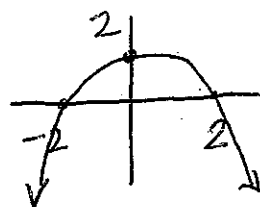
To calculate the volume of ice-cream, the area bounded by the section of the line $y = 4x$ between $(0,0)$ and $(1,4)$, the part of the parabola between $(1,4)$, $(0,5)$ and the y -axis, was rotated about the y -axis. Determine the total quantity of ice-cream contained in the cone and the scoop on top, in terms of π .

End of Question 4

End of Section B

End of Examination

① (a) $4 - x^2 \geq 0$
 $(2-x)(2+x) \geq 0$



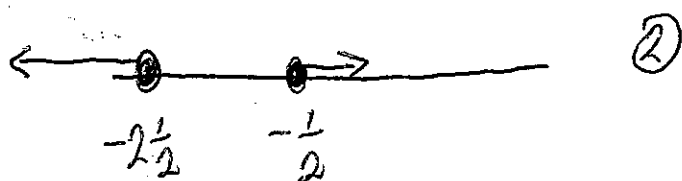
25

D $-2 \leq x \leq 2$. ①

* Range $0 \leq y \leq 4$ ←

b) $|2x+3| \geq 2$

$$\begin{aligned} 2x+3 &\geq 2 & 2x+3 &\leq -2 \\ 2x &\geq -1 & 2x &\leq -5 \\ x &\geq -\frac{1}{2} & x &\leq -2\frac{1}{2} \end{aligned}$$



c) $f(x) = 2ax + b$

$f(-1) = -2a + b = 5$

$f(2) = 4a + b = -1$

$-6a = 6$

$a = -1$ *

so $-4 + b = -1 \Rightarrow b = 3$ * ②

d) (i) $\frac{d}{dx} (6x^5 - 3x^3 - 7x + 1) = 30x^4 - 9x^2 - 7$ ①

(ii) $\frac{d}{dx} (7x^{-1}) = -7x^{-2} = \frac{-7}{x^2}$ ①

(iii) $\frac{d}{dx} (4(1-x^3)^5) = 20(1-x^3)^4 \times -3x^2$
 $= -60x^2(1-x^3)^4$ ②

$$d) \text{ (iv) } \frac{d}{dx} (\ln(6x^2-3)) = \frac{1}{6x^2-3} \times 12x$$

$$= \frac{12x}{6x^2-3} = \textcircled{2} \frac{4x}{2x^2-1}$$

$$(v) \frac{d}{dx} \left(\frac{x-1}{3x-4} \right) = \frac{(3x-4) \times 1 - (x-1) \times 3}{(3x-4)^2}$$

$$= \frac{3x-4-3x+3}{(3x-4)^2} = \frac{-1}{(3x-4)^2} \textcircled{2}$$

$$(vi) \frac{d}{dx} (xe^x) = xe^x + e^x \times 1$$

$$= e^x(x+1) \textcircled{2}$$

$$e) \text{ (i) } \int (x^2-2)^2 dx = \int (x^4 - 4x^2 + 4) dx$$

$$= \frac{x^5}{5} - \frac{4x^3}{3} + 4x + C \textcircled{2}$$

$$(ii) \int \frac{x^2-2}{x} dx = \int \frac{x^2}{x} - \frac{2}{x} dx$$

$$= \int x dx - 2 \int \frac{1}{x} dx$$

$$= \frac{x^2}{2} - 2 \ln x + C \textcircled{2}$$

$$(iii) \frac{1}{2} \int \frac{2x}{x^2-2} dx = \frac{1}{2} \ln(x^2-2) + C \textcircled{2}$$

Q2 (a) $\ln x = 2 \ln x$
 $2 \ln x - \ln x = 0$
 $\ln x = 0$

$\ln x = \ln x^2$
 $x = x^2$
 $x^2 - x = 0$
 $x(x-1) = 0$
 $x = 0, x = 1$
 $x = 1$ (not 0)

$\log_e x = 0$

$e = 1 = x$ (1)

(i) $\int_0^3 x^{\frac{1}{2}} dx = \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^3$

exact or approx
 $= \frac{2}{3} x \sqrt{x} \Big|_0^3$
 $= \frac{2}{3} \times 3 \sqrt{3}$
 $= 2\sqrt{3}$ (2)

(iii) $\int_1^{e-1} \frac{1}{x+1} dx = 2\sqrt{3} (3.4641...)$

$= \ln(x+1) \Big|_1^{e-1}$

$= \ln(e) - \ln(2)$

$= 1 - \ln 2$ (2)

(d) (i) $g(x) = \frac{2}{x^2-1}$

$g(-x) = \frac{2}{(-x)^2-1} = \frac{2}{x^2-1}$

$g(x) = g(-x)$ even fn. (2)

(b) $f(x) = e^{x-2}$

$f(-1) = e^{-3}$

$= 0.0498$ (1)

(4 DP)

(ii) $\int_0^3 \frac{1}{(2x-3)^2} dx$

$= \int_0^3 (2x-3)^{-2} dx$

$= \left[\frac{(2x-3)^{-1}}{-1 \times 2} \right]_0^3$

$= -\frac{1}{2} \left[\frac{1}{(2x-3)} \right]_0^3$

$= \left(-\frac{1}{2} \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{-3} \right)$

$= \left(-\frac{1}{6} - \frac{1}{6} \right) = -\frac{1}{3}$

(2)

(ii) $x^2 - 1 \neq 0$

$x^2 \neq 1$

$x \neq \pm 1$ (1)

12 (e) $L = r\theta$

$$10\pi = r \times \frac{\pi}{6}$$

$$r = \cancel{10\pi} \times \frac{6}{\cancel{\pi}}$$

$$= 60 \text{ cm.}$$

$$\begin{aligned} \text{shaded area} &= \frac{1}{2} r^2 (\theta - \sin \theta) \\ &= \frac{1}{2} \times 3600 \left(\frac{\pi}{6} - \sin 30^\circ \right) \\ &= 1800 \left(\frac{\pi}{6} - \frac{1}{2} \right) \\ &= 1800 \left(\frac{\pi - 3}{6} \right) \\ \text{exact} &= 300(\pi - 3) \text{ cm}^2 \quad (2) \end{aligned}$$

(f) (i) (A) when $\frac{dy}{dx} = 3(x-1)^2 = 0$
 $x = 1.$

LHS of $x=1$, $\frac{dy}{dx} > 0$

RHS of $x=1$, $\frac{dy}{dx} > 0$

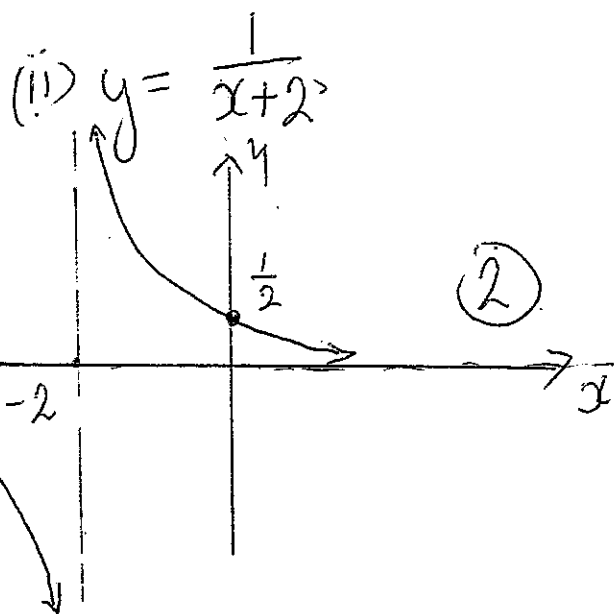
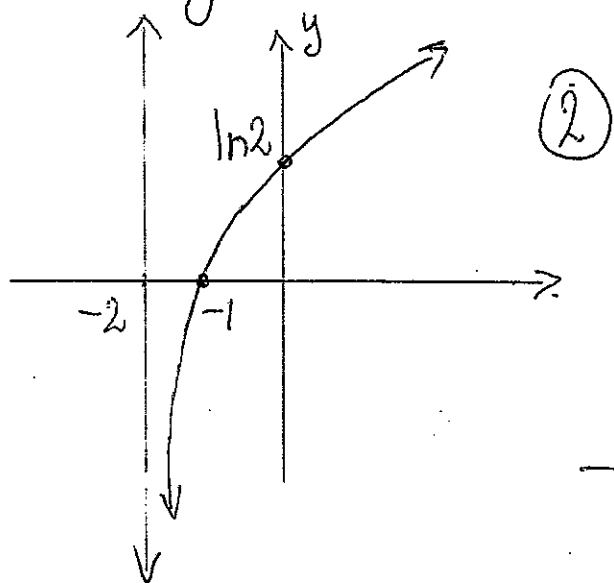
(B) except for $x=1$, $\frac{dy}{dx} > 0$ for all x . (1)

(C) at $x=1$, there is no max or min stat. pt
 but a stat pt of inflexion (1)

(f) (i) $y = \log(x+2)$

$$x+2 > 0$$

$$x > -2$$



Q2 (f) (ii)

x	0	1	2
sign of y''	-	0	+

(i)

(iii) because of the y'' sign change to the left and right of $x=1$, we have a point of inflexion. (i)

(iv) $y = \int 3(x-1)^2 dx$

$$y = \frac{3(x-1)^3}{3} + C$$

at (0,0) $0 = -1 + C$

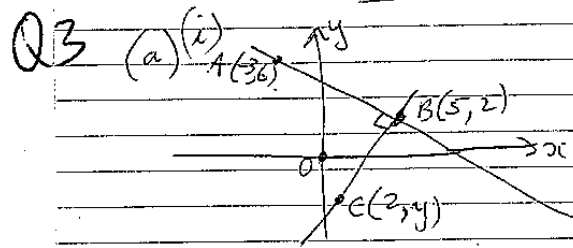
$$C = 1$$

$$y = (x-1)^3 + 1$$

(2)

$$y = x^3 - 3x^2 + 2x + 1$$

Section B Q3 (25)



(ii) $x + 2y = 9$

$A(-3, 6) \Rightarrow -3 + 12 = 9 \checkmark$

$B(5, 2) \Rightarrow 5 + 4 = 9 \checkmark$ A, B satisfy eqn
 $\therefore$ they lie on line.

(iii) $d_{AB} = \sqrt{(5+3)^2 + (2-6)^2}$
 $= \sqrt{64 + 16}$
 $= \sqrt{80} = 4\sqrt{5}$

(iv) Area $\triangle OAB$

Now perp. distance from O to AB
 i.e. $(0, 0)$ to $x + 2y = 9$

is $d = \frac{|10 + 0 - 9|}{\sqrt{1+4}} = \frac{1}{\sqrt{5}}$

(v) Then Area $\triangle OAB = \frac{1}{2}bh$
 $= \frac{1}{2} \times 4\sqrt{5} \times \frac{1}{\sqrt{5}}$
 $= 18 \text{ units}^2$

(vi) $C(2, y) = ?$ Now $m_{AB} = \frac{6-2}{-3-5} = \frac{4}{-8} = -\frac{1}{2} \checkmark$

Then $m_{BC} = \frac{y-2}{2-5} = \frac{y-2}{-3} = 2$

$\Rightarrow y-2 = -6$
 $y = -4$

$\therefore C = (2, -4)$

(vii) Line AC Passes through $A(-3, 6)$ and $C(2, -4)$
 Eqn of AC: $y - 6 = \frac{6+4}{-3-2}(x+3)$

$y - 6 = -2(x+3)$
 $y = -2x$

$\Rightarrow (0, 0) = 0 = -2 \times 0$
 $\therefore (0, 0)$ does lie on line AC
 as it satisfies eqn.

(viii) $D = (d, 0)$ ABCD is rect. and $C = (2, -4)$
 $\Rightarrow AB \parallel CD, AB \perp BC$

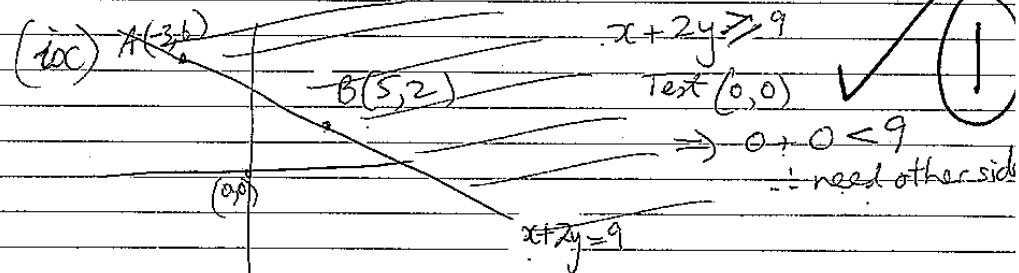
$m_{AB} = -\frac{1}{2}$ and $m_{BC} = 2$

Then $m_{AB} = m_{CD} \Rightarrow -\frac{1}{2} = \frac{-4}{d-2}$

$-d+2 = 8$

$d = -6$

$\therefore D = (-6, 0)$



9(b) $y = x \ln x$ (3)

$$y' = x \cdot \frac{1}{x} + \ln x = 1 + \ln x \quad \checkmark$$

At (e, e) $y' = 1 + \ln e$
 $y' = 2 = m$

Then gradient of normal $= -\frac{1}{2}$ $\checkmark$

At (e, e) , eqn of normal is

$$y - e = -\frac{1}{2}(x - e)$$

$$2y - 2e = -x + e$$

$$\Rightarrow x + 2y - 3e = 0 \quad \checkmark$$

c) $y = e^{3x}$

Parallel to line $y = 6x \Rightarrow m = 6 \quad \checkmark$

$$y' = 3e^{3x}$$

Then $3e^{3x} = 6$

$$e^{3x} = 2$$

$$3x = \ln 2$$

$$x = \frac{\ln 2}{3} \quad \checkmark$$

(d) (i) $y = 4 - x$ (4)

Find X when $y = 0$, $0 = 4 - x$
 $x = 4$

$$\therefore X = (4, 0) \quad \checkmark$$

Find A

$$y = 4 - x \quad (1)$$

$$y = 3x^2 \quad (2)$$

Sub (1) into (2)

$$\Rightarrow 3x^2 = 4 - x$$

$$3x^2 + x - 4 = 0$$

$$(3x + 4)(x - 1) = 0$$

$$x = -\frac{4}{3} \text{ or } x = 1$$

For A, $x, y > 0 \Rightarrow x = 1$. Sub into (2) $\Rightarrow y = 3$

$$\therefore A = (1, 3)$$

(ii) Area OABC $= \int_0^1 3x^2 dx + \int_1^4 (4 - x) dx + \int_4^5 (4 - x) dx$

$$= \left[x^3 \right]_0^1 + \left[4x - \frac{x^2}{2} \right]_1^4 + \left[4x - \frac{x^2}{2} \right]_4^5$$

$$= (1 - 0) + \left[(16 - 8) - \left(4 - \frac{1}{2} \right) \right] + \left[(20 - \frac{25}{2}) - (16 - 8) \right]$$

$$= 1 + 4\frac{1}{2} + 1\frac{1}{2}$$

$$= 6 \text{ square units.} \quad \checkmark$$

$$(e) y = (\ln x)^2 \quad (5)$$

$$y' = 2 \ln x \cdot \frac{1}{x} = \frac{2 \ln x}{x}$$

$$\text{Then } \int_1^2 \frac{\ln x}{x} dx = \frac{1}{2} \int_1^2 \frac{2 \ln x}{x} dx$$

$$= \frac{1}{2} \left[(\ln x)^2 \right]_1^2$$

$$= \frac{1}{2} \left((\ln 2)^2 - (\ln 1)^2 \right)$$

$$= \frac{1}{2} (\ln 2)^2$$

(3)

[Section B-Q4] (20) (1)

$$(a) \frac{d}{dx} (b^x) = b^x \ln b$$

$$\text{Then } \int_0^\pi \pi^x dx$$

$$= \frac{1}{\ln \pi} \int_0^\pi \pi^x \ln \pi dx$$

$$= \frac{1}{\ln \pi} \left[\pi^x \right]_0^\pi$$

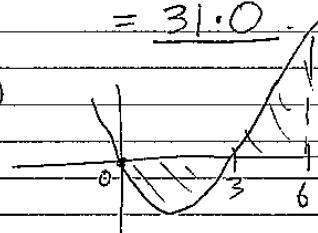
$$= \frac{1}{\ln \pi} [\pi^\pi - \pi^0]$$

$$= \frac{1}{\ln \pi} [\pi^\pi - 1]$$

$$= 30.978$$

$$= 31.0$$

(b)



$$y = x^2 - 3x$$

$$= x(x-3)$$

(a) This is incorrect because some of the area is below the x-axis so has a negatively signed area and some is above which has a positively signed area.

Finding $\int_0^6 f(x) dx$ finds the difference in these 2 areas.

$$(b) \text{Area} = \left| \int_0^3 (x^2 - 3x) dx \right| + \int_3^6 (x^2 - 3x) dx$$

$$= \left| \left[\frac{x^3}{3} - \frac{3x^2}{2} \right]_0^3 \right| + \left[\frac{x^3}{3} - \frac{3x^2}{2} \right]_3^6$$

Q4 (cont)

(b)(b) cont

$$\text{Area} = \left| \left(9 - \frac{27}{2} \right) - 0 \right| + \left(\frac{216}{3} - \frac{108}{2} \right) - \left(9 - \frac{27}{2} \right)$$

$$= 27 \text{ units}^2$$

c) $f(x) = \frac{\ln \sqrt{x}}{x} = \frac{\ln x^{\frac{1}{2}}}{x} = \frac{\frac{1}{2} \ln x}{x} = \frac{\ln x}{2x}$

Domain $x > 0$

(i) $f'(x) = \frac{2x \cdot \left(\frac{1}{x}\right) - \ln x \cdot 2}{4x^2}$

$$= \frac{2 - 2 \ln x}{4x^2}$$

$$f'(x) = \frac{1 - \ln x}{2x^2}$$

For stat pts, $f'(x) = 0 \Rightarrow 1 - \ln x = 0$
 $\ln x = 1$
 $\Rightarrow x = e$

When $x = e$, $y = \frac{1}{2e}$
 $\Rightarrow$ Stat pt at $(e, \frac{1}{2e})$

Type?

x	2	e	3
y	0.17	0	-0.05
	✓	-	✓

max at $(e, \frac{1}{2e})$

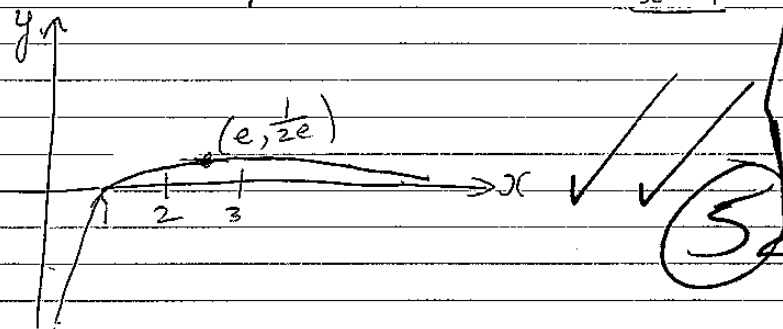
(ii) $x \neq 0, y \neq 0$ As $x \rightarrow \infty, y \rightarrow 0$

Critical Value at $x=0 \Rightarrow$ vertical asymptote

x	0	$\frac{1}{2}$
y	N.D.	$3.38 \rightarrow$
		↑ +ve

As $x \rightarrow 0, y \rightarrow -\infty$

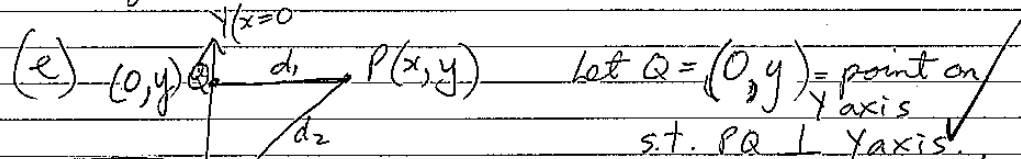
4(c)(ii) cont Intercept when $y=0 \Rightarrow \ln x = 0$
 $x = 1$



(d) $y = \log_2 x = \frac{\ln x}{\ln 2} = \frac{1}{\ln 2} \ln x$

$$y' = \frac{1}{\ln 2} \cdot \frac{1}{x}$$

$$y' = \frac{1}{x \ln 2}$$



Then $d_1^2 = d_2^2$

$$\Rightarrow x^2 = y^2 + (x-1)^2$$

Then $y^2 = x^2 - x^2 + 2x - 1$

$$y^2 = 2x - 1$$

$$y = 2\left(x - \frac{1}{2}\right)$$

3

(4)

4. (f) $V = \int \pi x^2 dy$ for rotation about Y-axis

$$\text{For } y = 4x \\ \Rightarrow x = \frac{y}{4}$$

$$\text{and for } y = 5 - x^2$$

$$x^2 = 5 - y \\ x = \sqrt{5 - y} \quad \text{since } x > 0$$

$$\text{Then } V = \int_0^4 \pi \frac{y^2}{16} dy + \int_4^5 \pi (5 - y) dy.$$

$$= \frac{\pi}{16} \left[\frac{y^3}{3} \right]_0^4 + \pi \left[5y - \frac{y^2}{2} \right]_4^5$$

$$= \frac{\pi}{16} \left[\frac{64}{3} - 0 \right] + \pi \left[\left(25 - \frac{25}{2} \right) - \left(20 - 8 \right) \right]$$

$$= \frac{4}{3} \pi + \frac{\pi}{2}$$

$$= \frac{11\pi}{6} \text{ cubic units}$$

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